```
lab-03-15@lab-03-15-OptiPlex-3280-AIO:~$ gcc sjf_pree_1288.c
lab-03-15@lab-03-15-OptiPlex-3280-AIO:~$ ./a.out
Enter number of processes: 5
Enter Arrival Time and Burst Time:
P1 AT: 6
P1 BT: 7
P2 AT: 2
P2 BT: 9
P3 AT: 2
P3 BT: 3
P4 AT: 1
P4 BT: 4
P5 AT: 2
P5 BT: 5

Gantt Chart:
| P4 | P3 | P4 | P5 | P1 | P2 |

Process AT      BT      CT      WT      TAT
P1      6       7       20      7       14
P2      2       9       29      18      27
P3      2       3       5       0       3
P4      1       4       8       3       7
P5      2       5       13      6       11

Average WT = 6.80
Average TAT = 12.40
lab-03-15@lab-03-15-OptiPlex-3280-AIO:~$
```

```c
#include <stdio.h>
#include <limits.h>
int main() {
    int n;
    printf("Enter number of processes: ");
    scanf("%d", &n);

    int at[n], bt[n], ct[n], wt[n], tat[n];
    int completed[n];

    for(int i = 0; i < n; i++)
        completed[i] = 0;

    printf("Enter Arrival Time and Burst Time:\n");
    for(int i = 0; i < n; i++) {
        printf("P%d AT: ", i+1);
        scanf("%d", &at[i]);
        printf("P%d BT: ", i+1);
        scanf("%d", &bt[i]);
    }

    int time = 0, done = 0;
    float total_wt = 0, total_tat = 0;

    printf("\nGantt Chart:\n|");

    while(done < n) {
        int idx = -1;
        int min_bt = INT_MAX;

        for(int i = 0; i < n; i++) {
            if(at[i] <= time && completed[i] == 0) {
                if(bt[i] < min_bt) {
                    min_bt = bt[i];
                    idx = i;
}
```

```
lab-03-15@lab-03-15-OptiPlex-3280-AIO:~$ vim sjf_Non_pree_1288.c
lab-03-15@lab-03-15-OptiPlex-3280-AIO:~$ gcc sjf_Non_pree_1288.c
lab-03-15@lab-03-15-OptiPlex-3280-AIO:~$ ./a.out
Enter number of processes: 5
Enter Arrival Time and Burst Time:
P1 AT: 6
P1 BT: 7
P2 AT: 2
P2 BT: 9
P3 AT: 2
P3 BT: 3
P4 AT: 1
P4 BT: 4
P5 AT: 2
P5 BT: 5

Gantt Chart:
| P4 | P3 | P5 | P1 | P2 |
0    20    29    8    5    13

Process AT      BT      CT      WT      TAT
P1      6       7       20      7       14
P2      2       9       29      18      27
P3      2       3       8       3       6
P4      1       4       5       0       4
P5      2       5       13      6       11

Average WT = 6.80
Average TAT = 12.40
lab-03-15@lab-03-15-OptiPlex-3280-AIO:~$
```

```
        }
    }
    if(idx != -1) {
        if(prev != idx) {
            printf(" P%d |", idx+1);
            prev = idx;
        }
        rt[idx]--;
        time++;
        if(rt[idx] == 0) {
            completed++;
            ct[idx] = time;
            tat[idx] = ct[idx] - at[idx];
            wt[idx] = tat[idx] - bt[idx];
            total_wt += wt[idx];
            total_tat += tat[idx];
        }
    } else {
        time++;
    }
}
printf("\n\nProcess\tAT\tBT\tCT\tWT\tTAT\n");
for(int i = 0; i < n; i++)
    printf("P%d\t%d\t%d\t%d\t%d\t%d\n",
        i+1, at[i], bt[i], ct[i], wt[i], tat[i]);
printf("\nAverage WT = %.2f", total_wt/n);
printf("\nAverage TAT = %.2f\n", total_tat/n);

    return 0;
}
```

```c
                    idx = i;
                }
            }
        }

        if(idx != -1) {
            printf(" P%d |", idx+1);
            time += bt[idx];
            ct[idx] = time;
            tat[idx] = ct[idx] - at[idx];
            wt[idx] = tat[idx] - bt[idx];

            total_wt += wt[idx];
            total_tat += tat[idx];

            completed[idx] = 1;
            done++;
        } else {
            time++; // idle
        }
    }

    printf("\n0");
    for(int i = 0; i < n; i++)
        printf("    %d", ct[i]);

    printf("\n\nProcess\tAT\tBT\tCT\tWT\tTAT\n");
    for(int i = 0; i < n; i++)
        printf("P%d\t%d\t%d\t%d\t%d\t%d\n",
               i+1, at[i], bt[i], ct[i], wt[i], tat[i]);

    printf("\nAverage WT = %.2f", total_wt/n);
    printf("\nAverage TAT = %.2f\n", total_tat/n);

    return 0;
}
```

```c
#include <stdio.h>
#include <limits.h>

int main() {
    int n;
    printf("Enter number of processes: ");
    scanf("%d", &n);

    int at[n], bt[n], rt[n], ct[n], wt[n], tat[n];

    printf("Enter Arrival Time and Burst Time:\n");
    for(int i = 0; i < n; i++) {
        printf("P%d AT: ", i+1);
        scanf("%d", &at[i]);
        printf("P%d BT: ", i+1);
        scanf("%d", &bt[i]);
        rt[i] = bt[i];
    }

    int time = 0, completed = 0;
    int prev = -1;
    float total_wt = 0, total_tat = 0;

    printf("\nGantt Chart:\n|");

    while(completed < n) {
        int idx = -1;
        int min_rt = INT_MAX;

        for(int i = 0; i < n; i++) {
            if(at[i] <= time && rt[i] > 0 && rt[i] < min_rt) {
                min_rt = rt[i];
                idx = i;
            }
        }

        if(idx != -1) {
```